Research Report

System Analysis of Uber – Understanding a Real-Time Matching Platform

Objective of the Research

The purpose of this research is to understand how Uber works as a real-time location-based platform and identify the software architecture, workflow, and algorithms that can be adapted for developing a Blood Donation Management System.

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1. Introduction

Uber is commonly known as a ride-booking application, but from a software engineering perspective, it is a **real-time location-based matching platform**.

Its primary objective is not simply booking taxis but efficiently matching the most suitable driver with a passenger based on multiple real-time factors such as location, estimated arrival time (ETA), availability, traffic conditions, and driver status.

This makes Uber an excellent case study for any application that needs intelligent matching between two different users.

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2. Core Objective of Uber

The main objective of Uber is:

* Connect passengers with nearby drivers.

* Reduce passenger waiting time.

* Select the most suitable driver instead of only the nearest driver.

* Track both passenger and driver in real time.

* Provide a secure, verified, and seamless journey.

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3. Major Components of Uber

Uber consists of multiple independent modules that work together.

a) Authentication System

* User registration and login

* Mobile number verification using OTP

* Session management

* Secure user authentication

Purpose:

Ensures that only verified users can access the platform.

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b) User Profile Management

Stores user information including:

* Name

* Mobile number

* Saved locations

* Payment methods

* Ride history

* Preferences

Purpose:

Provides a personalized experience and stores user-related data.

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c) GPS and Location Service

This is one of the most important modules.

The application continuously collects:

* Latitude

* Longitude

* Speed

* Direction

* Timestamp

Both passenger and driver send live location updates to Uber's servers.

Purpose:

Allows accurate tracking and real-time navigation.

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d) Driver Availability Management

Every driver has a live status.

Possible statuses:

* Online
* Offline
* Busy
* On Trip
* Available

The server continuously updates this information.

Purpose:
Ensures that only available drivers are considered for ride requests.

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e) Location Search Mechanism

Uber does not compare a passenger with every driver in the city.

Instead, the city is divided into smaller geographical regions (cells or grids).

When a ride request is created:

* The passenger's location is identified.
* Only drivers present in the same or nearby regions are searched.

Purpose:

Reduces search time and improves scalability.

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f) Route Optimization

After entering the destination, Uber calculates:

* Shortest route
* Fastest route
* Estimated travel distance
* Estimated travel time (ETA)

Traffic conditions are also considered.

Purpose:

Provides accurate arrival time and route suggestions.

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g) Pricing Engine

Uber's fare is calculated using multiple parameters:

* Base fare
* Distance travelled
* Time taken
* Traffic conditions
* Demand and supply (Surge Pricing)
* Toll charges
* Taxes

Purpose:

Generate a fair and dynamic fare estimate.

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h) Matching Engine

The Matching Engine is the core of the Uber platform.

It selects the best driver using multiple criteria instead of choosing only the nearest driver.

Possible matching parameters include:

* Distance from passenger
* Estimated arrival time
* Driver availability
* Acceptance probability

* Driver rating
* Current traffic

The system assigns a priority score and selects the most suitable driver.

Purpose:

Improve efficiency and reduce waiting time.

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i) Real-Time Communication

Once a driver accepts the request:

* Passenger location is shared.
* Driver location is updated continuously.
* Estimated arrival time changes dynamically.
* Notifications are exchanged instantly.

Purpose:

Maintain synchronization between passenger and driver.

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j) Ride Verification

In many regions, Uber uses OTP verification before the trip starts.

Purpose:

* Prevent unauthorized rides.

* Verify passenger identity.

* Improve security.

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k) Trip Monitoring

During the journey Uber records:

* GPS coordinates

* Route followed

* Speed

* Time

* Distance

* Stops

Purpose:

Safety, analytics, customer support, and dispute resolution.

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l) Payment System

After ride completion:

* Final fare is calculated.

* Payment is processed.

* Digital receipt is generated.

* Transaction history is updated.

Purpose:

Provide secure and transparent payment.

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m) Rating and Feedback System

Both passenger and driver rate each other.

Purpose:

Maintain service quality and identify poor-performing users.

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4. Overall Workflow

1. User opens the application.
2. Authentication is verified.
3. GPS location is obtained.
4. Nearby available drivers are identified.
5. Passenger enters destination.
6. Route and fare are calculated.
7. Ride request is created.

8. Matching Engine selects the best driver.

9. Ride request is sent to the driver.

10. Driver accepts the request.

11. Live tracking begins.

12. OTP verification is completed (if applicable).

13. Ride starts.

14. Ride ends.

15. Payment is completed.

16. Ratings and trip history are stored.

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5. Key Technologies Used

* GPS for location tracking

* Digital maps and navigation

* Real-time communication

* Cloud servers

* Databases

* Matching algorithms

* Route optimization

* Dynamic pricing

* Push notification services

* Authentication and OTP verification

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6. Research Findings

The analysis shows that Uber is not simply a taxi-booking application. It is a highly optimized real-time matching platform consisting of multiple interconnected systems such as authentication, GPS tracking, intelligent matching, route optimization, real-time communication, verification, payment, and analytics.

The most important component is the Matching Engine, which uses multiple parameters to select the best available driver instead of merely choosing the nearest one. This architecture can be adapted to many domains where intelligent real-time matching between two parties is required.

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7. Relevance to Our Blood Donation Project

This research is highly relevant because a Blood Donation Management System also requires intelligent matching between two different users.

The same architectural concepts used in Uber can be adapted as follows:

* Passenger → Blood Recipient
* Driver → Blood Donor
* Ride Request → Blood Request
* Driver Matching → Donor Matching
* Live Tracking → Donor Arrival Tracking
* OTP Verification → Donation Verification
* Navigation → Hospital Navigation

Instead of matching passengers with drivers, our system will match recipients with the most suitable eligible blood donors based on blood compatibility, location, availability, and medical eligibility.

This research provides a strong foundation for designing the system architecture, workflow, and matching algorithm of our proposed Blood Donation Management Platform.

CHAPTER 1

AI/ML in Blood Inventory Management

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are transforming healthcare by enabling data-driven decision-making. In blood inventory management, AI is used to predict blood demand, optimize inventory, reduce blood wastage, recommend suitable donors, and support hospitals during emergencies.

Blood is a **perishable biological resource**, meaning it has a limited storage life and cannot be manufactured artificially. Therefore, maintaining the right quantity of each blood group at the right place and the right time is one of the biggest challenges for hospitals and blood banks.

Traditional inventory systems rely mainly on manual monitoring and historical averages. These methods often fail to adapt to sudden emergencies, seasonal variations, disease outbreaks, or unexpected increases in demand. AI and ML can overcome these limitations by analyzing large volumes of historical and real-time data to generate accurate predictions and intelligent recommendations.

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2. Problem Statement

Blood banks face several operational challenges that directly affect patient care.

Major Problems

A. Blood Shortage

Hospitals may not have enough blood units during emergencies such as:

* Road accidents
* Major surgeries
* Natural disasters
* Pregnancy complications
* Cancer treatments

Result:

Patients may not receive blood in time.

B. Blood Wastage

Blood cannot be stored indefinitely.

Each blood component has a different shelf life:

Blood Component	Approximate Shelf Life
Red Blood Cells	42 days
Platelets	5–7 days

| Plasma | Up to 1 year (when frozen) |

If blood is not used before expiry, it must be discarded.

Result:

Loss of valuable medical resources.

C. Uneven Inventory Distribution

Example:

Blood Bank	O+ Units
Blood Bank A	6
Blood Bank B	180
Blood Bank C	4

One blood bank experiences a shortage while another has excess stock.

Result:

Poor resource utilization.

D. Manual Decision Making

Most hospitals depend on human judgment for:

* Inventory planning
* Stock replenishment
* Emergency response

Manual decisions become difficult when thousands of blood units and multiple hospitals are involved.

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3. Why Traditional Systems Are Not Enough

Traditional systems generally provide only basic information such as:

* Current stock
* Blood group
* Expiry date

However, they cannot answer important questions such as:

* Which blood group will be in shortage next week?
* Which blood bags should be used first?
* Which hospital nearby has surplus blood?
* Which donor is most likely to respond quickly?
* Should an emergency blood donation camp be organized today?

These questions require prediction and intelligent decision-making, which is where AI becomes valuable.

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4. Why AI/ML is Needed

Artificial Intelligence enables hospitals to move from **reactive management** to **predictive management**.

Instead of waiting for a shortage to occur, AI predicts future demand and recommends preventive actions.

AI continuously learns from historical data and identifies patterns that may not be visible through manual observation.

For example:

* Increase in accident cases during festivals
* Higher blood demand during monsoon
* Seasonal variations in donations
* Increased demand in cancer treatment centers

These patterns help hospitals prepare in advance.

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5. Data Collection

Machine Learning models require large amounts of high-quality data.

The following datasets can be collected.

A. Blood Inventory Data

* Blood Group
* Blood Component
* Collection Date
* Expiry Date
* Current Stock
* Storage Temperature
* Blood Bag ID

Purpose:

To understand current inventory status.

B. Hospital Data

* Number of surgeries
* Emergency admissions
* ICU patients
* Trauma cases
* Daily blood consumption

Purpose:

To estimate future blood demand.

C. Donor Data

* Blood Group
* Age
* Gender
* Weight
* Previous Donation Date
* Medical Eligibility
* Current Location
* Availability Status

Purpose:

To identify suitable donors.

D. External Data

AI can also use external factors.

Examples:

* Festivals

* Public holidays

* Weather

* Disease outbreaks

* Road accident statistics

* Population density

Purpose:

Improve prediction accuracy.

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6. Data Preprocessing

Raw data cannot be directly used for Machine Learning.

It must first be cleaned and prepared.

The preprocessing process consists of the following steps.

A. Missing Value Handling

Some records may contain incomplete information.

Example:

Donation Date = Missing

Solution:

* Remove incomplete records
  or
* Estimate missing values

B. Duplicate Removal

Sometimes the same donor registers multiple times.

Duplicate records are removed to improve accuracy.

C. Data Standardization

Different hospitals may store information differently.

Example:

O Positive

O+

O POS

All are converted into

O+

This ensures consistency.

D. Feature Engineering

Feature Engineering means creating new useful information from existing data.

Example

Collection Date

↓

Days Until Expiry

Last Donation Date

↓

Days Since Last Donation

Date

↓

Festival Season

These newly created features improve prediction accuracy.

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7. Machine Learning Pipeline

The AI system follows a structured workflow.

```text
Data Collection
    ↓
Data Cleaning
    ↓
Data Preprocessing
    ↓
Feature Engineering
    ↓
Model Training
    ↓
Prediction
```

↓

Decision Support

↓

Hospital Dashboard

```

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# 8. AI Module 1 – Blood Demand Forecasting

## Objective

Predict future blood demand before shortages occur.

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### Input Data

* Historical blood requests
* Hospital admissions
* Blood group demand
* Surgeries
* Festivals
* Weather
* Road accidents
* Disease outbreaks

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```

Suitable Machine Learning Models

* Prophet
* ARIMA
* LSTM
* XGBoost Regression

Output

Example

Blood Group	Predicted Demand
O+	180 Units
A+	120 Units
B+	95 Units
O−	35 Units

Benefits

* Prevent shortages
* Better planning
* Reduced emergency situations

Research Observation

Demand forecasting is one of the most important applications of AI because hospitals can prepare inventory **before** demand increases instead of reacting after the shortage occurs.

CHAPTER 2

Blood Demand Forecasting using Artificial Intelligence & Machine Learning

2.1 Introduction

Blood demand forecasting is the process of predicting the future requirement of different blood groups before shortages occur. It is one of the most important applications of Artificial Intelligence (AI) and Machine Learning (ML) in healthcare logistics because blood cannot be manufactured and has a limited shelf life.

Traditionally, hospitals estimate future blood demand based on previous experience or average monthly consumption. However, such methods are often inaccurate because blood demand changes due to accidents, surgeries, seasonal diseases, festivals, natural disasters, and public events.

AI-based forecasting systems analyze historical and real-time data to identify hidden patterns and generate accurate predictions. These predictions help hospitals prepare inventory in advance, reducing shortages, minimizing wastage, and improving emergency response.

2.2 Why is Blood Demand Forecasting Important?

A hospital cannot wait until blood runs out before arranging new donations.

For example,

Suppose today's stock is:

Blood Group	Current Stock
O+	150 Units
A+	80 Units
B+	70 Units
O−	12 Units

Without prediction, the hospital only knows the current inventory.

AI answers a much more important question:

"How much blood will be required tomorrow, next week, or next month?"

This enables proactive planning instead of reactive management.

2.3 Problems with Traditional Forecasting

Traditional blood inventory management mainly depends on manual estimation and historical averages.

Limitations

- Assumes future demand will be similar to previous months.
- Cannot detect sudden increases in emergency cases.
- Ignores seasonal variations.

Does not consider local events.

Cannot learn from continuously changing data.

Often results in either overstocking or shortages.

Example

Suppose a hospital normally receives 50 O+ requests per week.

During a large festival:

Road accidents increase.

Actual demand becomes:

90 O+ requests

Traditional estimation still predicts:

50 requests

Result:

Blood shortage

Emergency donor search

Delay in patient treatment

2.4 Role of AI in Demand Forecasting

Artificial Intelligence continuously learns from historical and real-time data.

Instead of using one average value,

AI identifies complex relationships between multiple factors.

Example:

Festival

↓

Road Traffic Increases

↓

Accidents Increase

↓

Blood Demand Increases

Humans may not notice these hidden relationships, but Machine Learning models can identify them automatically.

2.5 Data Required for AI Forecasting

A Machine Learning model is only as good as the data it receives.

The following datasets can improve prediction accuracy.

A. Historical Blood Requests

This is the most important dataset.

It includes:

Blood Group

Number of Requests

Date

Hospital

Emergency Level

Example:

Date	Blood Group	Requests
1 Jan	O+	18
2 Jan	O+	22
3 Jan	O+	19

Purpose:

Helps identify demand patterns over time.

B. Hospital Operational Data

Hospitals generate large amounts of useful information.

Examples:

Number of surgeries

ICU admissions

Trauma patients

Cancer treatments

Maternity cases

Dialysis procedures

Purpose:

Estimate expected blood consumption.

C. Seasonal Data

Demand changes with seasons.

Examples:

Monsoon

Summer

Winter

Reason:

Different seasons affect accident rates and disease prevalence.

D. Festival and Public Event Data

Examples:

Diwali

Holi

Durga Puja

New Year

Cricket tournaments

Political rallies

Large gatherings may increase emergency cases or affect donor availability.

E. Road Accident Statistics

If accident frequency increases,

blood demand usually increases.

Data Sources:

Traffic Department

Government reports

Hospital emergency records

F. Disease Outbreak Data

Certain diseases increase blood requirements.

Examples:

Dengue (platelets)

Thalassemia (regular transfusions)

Cancer

Severe anemia

2.6 Data Preprocessing

Raw hospital data is often inconsistent.

Before training the model,

the data must be cleaned.

Step 1 – Remove Missing Values

Example:

Blood Group = O+

Request Count = Missing

Possible solutions:

Remove the record

Estimate the missing value

Replace with average

Step 2 – Remove Duplicate Records

Duplicate entries may occur if the same request is recorded multiple times.

Removing duplicates prevents biased predictions.

Step 3 – Standardize Data

Different hospitals may store data differently.

Example:

O Positive

O POS

O+

O Positive Blood

All are converted to:

O+

Step 4 – Feature Engineering

Machine Learning performs better when meaningful features are created.

Examples:

Original Data:

Collection Date

↓

New Feature:

Days Until Expiry

Original Data:

Date

↓

Day of Week

Month

Festival Season

These engineered features improve forecasting accuracy.

2.7 Machine Learning Models Used

Different forecasting problems require different models.

A. ARIMA (AutoRegressive Integrated Moving Average)

Purpose

Forecast future values using historical trends.

Suitable for:

Stable demand patterns

Small datasets

Advantages:

Easy to interpret

Fast

Works well for linear time-series

Limitations:

Cannot easily capture complex seasonal effects.

B. Facebook Prophet

Purpose:

Forecast seasonal healthcare data.

Advantages:

Handles holidays automatically.

Works well with missing values.

Easy to train.

Example:

Predict blood demand during Diwali or Christmas.

C. LSTM (Long Short-Term Memory Network)

LSTM is a deep learning model designed for sequential data.

Advantages:

Learns long-term patterns.

Handles complex time-series.

Suitable for large datasets.

Example:

Demand prediction over several years.

Limitation:

Requires more data and computing power.

D. XGBoost Regression

Unlike time-series-only models,

XGBoost can use multiple input variables.

Example Inputs:

Weather

Festivals

Accidents

Hospital admissions

Population density

Advantages:

High prediction accuracy

Handles mixed data types

Works well on structured tabular data

2.8 Model Training Process

Historical Data

↓

Data Cleaning

↓

Feature Engineering

↓

Train ML Model

↓

Model Validation

↓

Prediction

↓

Hospital Dashboard

The model continuously improves as new data becomes available.

2.9 Example Prediction

Suppose the AI predicts:

Blood Group	Current Stock	Predicted Weekly Demand	Status
O+	150	165	Prepare Additional Stock
A+	80	60	Sufficient
B+	70	95	Possible Shortage
O−	12	25	Critical

Instead of only showing numbers,

the AI provides actionable recommendations.

2.10 Decision Support

Based on predictions,

the system can automatically recommend:

If demand is expected to increase:

Organize blood donation camp.

Notify eligible donors.

Request stock from nearby blood banks.

Increase hospital inventory.

If demand is expected to decrease:

Avoid unnecessary collection.

Optimize storage.

Reduce wastage.

2.11 Advantages of AI-Based Forecasting

Predicts shortages before they occur.

Improves emergency preparedness.

Reduces blood wastage.

Optimizes donor recruitment.

Supports better hospital planning.

Reduces operational costs.

Enhances patient care.

2.12 Limitations

Requires high-quality historical data.

Accuracy depends on data availability.

Sudden disasters remain difficult to predict perfectly.

Models require regular retraining with new data.

Human experts should always validate AI recommendations.

2.13 Research Observation

Blood demand forecasting is not simply a prediction problem; it is a decision-support system. The objective is not only to estimate future demand but also to recommend actions that improve inventory availability while minimizing wastage. AI enables hospitals to transition from reactive inventory management to proactive, data-driven planning.

2.14 Research Conclusion

The study concludes that Machine Learning can significantly improve blood inventory management by accurately forecasting future demand using historical records, hospital operations, seasonal trends, and external factors. By integrating predictive analytics into hospital workflows, healthcare institutions can reduce shortages, minimize wastage, improve donor engagement, and make faster, evidence-based decisions during emergencies.

Research Idea 1: Blood Digital Twin (Very Advanced)

Problem

Current AI only predicts.

It doesn't simulate.

Idea

Create a Digital Twin of every blood bank.

A Digital Twin is a virtual copy of the real blood inventory.

Whenever something happens in the real world,

the digital model updates instantly.

Example

Hospital A

Current Stock

O+ : 52

A+ : 34

B+ : 28

↓

One emergency surgery

↓

Digital Twin Updates

↓

O+ : 49

Now AI can run simulations.

Example

"What happens if tomorrow there are 40 accident cases?"

The Digital Twin answers

O− becomes critical.

Recommendation:

Transfer blood from Hospital C.

Instead of reacting,

the hospital prepares before the emergency.

Research Idea 2: Blood Demand Heatmap

Not everyone needs blood equally.

Different areas of a city have different demand.

AI creates a heatmap.

🔴 High Demand

Hospital Zone

🟡 Medium

Residential Area

🟢 Low

Industrial Area

The system recommends

"Organize the next blood camp here."

Research Idea 3: Donation Probability Score

Current apps notify everyone.

Bad idea.

AI predicts

Who will actually donate.

Example

Donor	Probability
Rahul	94%

Priya 82%

Aman 15%

Only notify the top donors.

Result

Less spam.

Higher success rate.

Research Idea 4: AI Donation Calendar

Instead of saying

Donate whenever possible.

AI predicts

Hospital demand

↓

Next Month

↓

High

↓

Organize Camp

on

18 October

Now donation camps become planned scientifically.

Research Idea 5: Blood Supply Chain Graph

Imagine every hospital is a node.

Hospital A

———

Hospital B

———

Hospital C

———

Hospital D

Every edge stores

travel time

transport cost

available stock

Now AI finds

the fastest blood movement.

This uses Graph Theory.

Research Idea 6: Explainable AI (XAI)

Hospitals don't trust black-box AI.

Instead of saying

Prediction

Need 250 O+

The system explains

Reason

+ 35 surgeries

+ Dengue cases increasing

+ Festival

Prediction Confidence

94%

Doctors trust AI more because it explains itself.

Research Idea 7: Hospital Risk Score

Every hospital gets a live score.

Hospital A

Risk

82%

🔴 High

Reason

Low O−

High Surgery Count

Now governments can monitor all hospitals from one dashboard.

Research Idea 8: AI Blood Ecosystem

Instead of managing only blood,

manage the entire ecosystem.

Donor

↓

Blood Bank

↓

Hospital

↓

Patient

↓

Recovery

↓

Eligible Again

↓

Donor

AI learns from the complete cycle.

Research Idea 9: Reinforcement Learning

This is PhD-level.

Normally AI predicts.

Reinforcement Learning learns by trial and error.

Example

AI asks

Should I

Transfer Blood

OR

Notify Donors

It receives feedback.

If shortages reduce,

Reward +10

If blood expires,

Penalty -10

Over months,

AI learns the best strategy automatically.

Research Idea 10 (This is my favourite)

National Blood Intelligence Network

Instead of every hospital using separate AI,

all hospitals contribute anonymous inventory data.

One national AI predicts

Odisha

O− shortage

Expected

Next Week

↓

Transfer

From

West Bengal

Before Crisis

This changes the system from hospital-level optimization to country-level optimization.